RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018-2019 (Odd Semester)

Innovative Practices Description

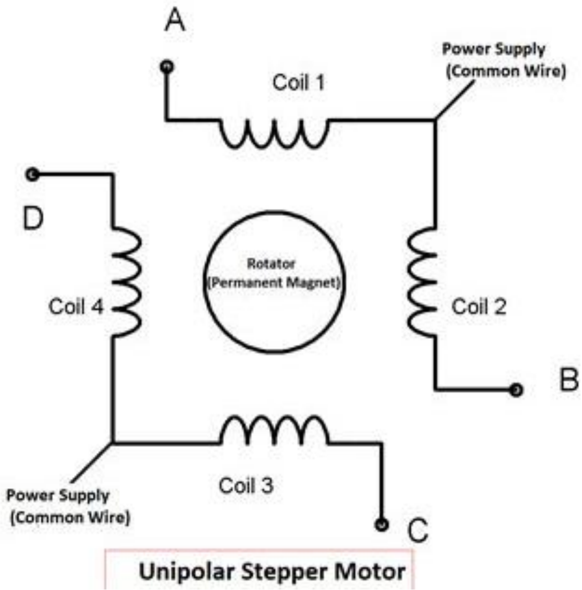
Degree, Semester& Branch: V Semester B.E. ECE A
Course Code & Title: EC6504 Microprocessor and Microcontroller
Name of the Faculty member: Mr.A.Azhagu Jaisudhan Pazhani
Name of the Topic: Stepper Motor Control
Name of the Innovative Practice: Real Time Examples

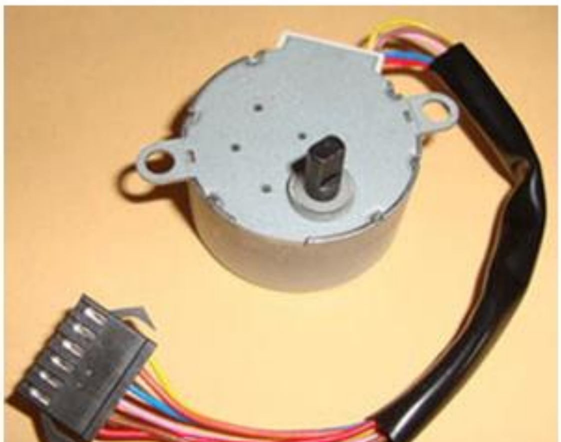
UNIT III TRAFFIC LIGHT SYSTEM

Description:

Stepper motor is brushless DC motor, which can be rotated in small angles, these angles are called steps. Generally stepper motor use 200 steps to complete 360 degree rotation, means its rotate 1.8 degree per step. Stepper motor used in many devices which needs precise rotational movement like robots, antennas, hard drives etc. We can rotate stepper motor to any particular angle by giving it proper instructions.

Stepper motors are basically two types: Unipolar and Bipolar. **Unipolar stepper** motor generally has five or six wire, in which four wires are one end of four stator coils, and other end of the all four coils is tied together which represents fifth wire, this is called common wire (common point). Generally there are two common wire, formed by connecting one end of the two-two coils as shown in below figure. Unipolar stepper motor is very common and popular because of its ease of use.





In **Bipolar stepper** motor there is just four wires coming out from two sets of coils, means there are no common wire. Stepper motor is made up of a stator and a rotator. Stator represents the four electromagnet coils which remain stationary around the rotator, and rotator represents permanent magnet which rotates. Whenever the coils energised by applying the current, the electromagnetic field is created, resulting the rotation of rotator (permanent magnet).

Coils should be energised in a particular sequence to make the rotator rotate. On the basis of this “sequence” we can divide the working method of **Unipolar stepper motor** in three modes: Wave drive mode, full step drive mode and half step drive mode.

Wave drive mode: In this mode one coil is energised at a time, all four coil are energised one after another. It produces less torque in compare with Full step drive mode but power consumption is less. Following is the table for producing this mode using microcontroller, means we need to give Logic 1 to the coils in the sequential manner.

Full Drive mode: In this, two coil are energised at the same time producing high torque. Power consumption is higher. We need to give Logic 1 to two coils at the same time, then to the next two coils and so on.

Steps	**A**	**B**	**C**	**D**
1	1	1	0	0
2	0	1	1	0
3	0	0	1	1
4	1	0	0	1

Interfacing with 8051 is very easy we just need to give the 0 and 1 to the four wires of stepper motor according to the above tables depending on which mode we want to run the stepper motor. And rest two wires should be connected to a proper 12v supply (depending on the stepper motor).

Course Outcome	**PO1**	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**
CO5	3	2	1	1	1	-	-	1	1	1	1	1

CO5: The student will be able to design and implement 8051 microcontroller based system.

References:

1. SK Mandal, “Microprocessor and Microcontroller”, Tata Mc Graw Hill Education.

2. NagoorKani, “Microprocessor and Microcontroller” Tata Mc Graw Hill Education.